

Structured Query Language



OUTER
LIKE
WHERE IN UPDATE DELETE
OR VALUES INTO JOIN
SELECT
INSERT
TABLE INNER TOP AS
AND FROM LIMIT DROP
BETWEEN



CREATE A TABLE

```
CREATE TABLE Character (Id int, Name varchar(30), Age int)
```

Character	Id	Name	Age

double ✗

float ✓



INSERT RECORDS

⚠ this format breaks if you later add or reorder columns!

```
INSERT INTO Character VALUES (1, 'Arya', 11)
```

```
INSERT INTO Character VALUES (1, 11, 'Arya')
```



Character	Id	Name	Age
	1	Arya	11



INSERT RECORDS

```
INSERT INTO Character VALUES (1, 'Arya', 11)
```

```
INSERT INTO Character (Id, Name, Age) VALUES (1, 'Arya', 11)
```

Character	Id	Name	Age
	1	Arya	11



ADD AND DELETE COLUMNS

```
ALTER TABLE Character DROP COLUMN Name
```

Character	Id	Age
	1	11



ADD AND DELETE COLUMNS

```
ALTER TABLE Character ADD Name varchar(30)
```

Character	Id	Age	Name
	1	11	null



ADD AND DELETE COLUMNS

```
ALTER TABLE Character ADD House varchar(20)
```

Character	Id	Age	Name	House
	1	11	null	null



ADD AND DELETE COLUMNS

```
INSERT INTO Character VALUES (1, 'Arya', 11)
```

⚠️ won't work unless
you swap the name
and age values!

Character	Id	Age	Name	House
	1	11	null	null



UPDATE A SPECIFIC ROW

```
UPDATE Character SET Name = 'Arya' WHERE Id = 1
```

```
UPDATE Character SET House = 'Stark' WHERE Id = 1
```

Character	Id	Age	Name	House
	1	11	Arya	Stark



UPDATE ALL ROWS

⚠ careful!
without
WHERE,
every single
row gets
updated!

```
UPDATE Character SET Name = 'Arya'
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Arya	Stark
	3	64	Arya	Lannister
	4	21	Arya	Targaryen



DELETE ROWS

```
DELETE FROM Character
```

```
DELETE FROM Character WHERE Id > 1
```

a best practice is to write it using the select keyword and then replace select with delete!

Character	Id	Age	Name	House
	1	11	Arya	Stark



SELECT ALL ROWS AND COLUMNS

```
SELECT * FROM Character
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister



SELECT SPECIFIC COLUMNS

```
SELECT Name, Age FROM Character ORDER BY Age ASC
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Name	Age
	Arya	11
	Sansa	13
	Daenerys	16
	Tyrion	32
	Cersei	36



SELECT SPECIFIC ROWS

```
SELECT Name, House FROM Character WHERE Id < 3
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Name	House
	Arya	Stark
	Tyrion	Lannister



AND / OR OPERATORS

```
SELECT Name FROM Character WHERE Id <= 3 AND House = 'Stark'
```

```
SELECT Name FROM Character WHERE Id = 1 OR Age = 13
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Name	
	Arya	
	Sansa	



IN / BETWEEN OPERATORS

```
SELECT Name, House FROM Character WHERE Id IN (1, 3, 4)
```

```
SELECT Name, House FROM Character WHERE Age BETWEEN 10 AND 25
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Name	House
	Arya	Stark
	Sansa	Stark
	Daenerys	Targaryen



LIKE OPERATOR

```
SELECT Id, Name FROM Character WHERE Name LIKE '%S%'
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Id	Name
	3	Sansa
	4	Daenerys
	5	Cersei



LIKE OPERATOR

```
SELECT Id, Name FROM Character WHERE Name LIKE 'A__a'
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Id	Name
	1	Arya



SQL FUNCTIONS

COUNT: returns the number of rows

SUM: returns the sum of a numeric column

AVG: returns the average of a set of values

MIN/MAX: gets the min/max value from a column

·
·

```
SELECT MAX(Age) FROM Character
```

MAX(Age)

36

```
SELECT COUNT(*) FROM Character
```

COUNT()

5



NESTED SELECTS

```
SELECT Name, House FROM Character WHERE Age = (  
    SELECT MIN(Age) FROM Character  
)
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Name	House
	Arya	Stark



JOINING TABLES

```
SELECT Appearance.Season, Character.Name
FROM Character INNER JOIN Appearance
ON Character.Id = Appearance.Character
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Season	Name
	1	Cersei
	2	Tyrion
	3	Arya
	4	Cersei
	5	Tyrion

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	2



USING ALIASES

```
SELECT a.Season, c.Name FROM Character AS c
       INNER JOIN Appearance AS a
       ON c.Id = a.Character
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

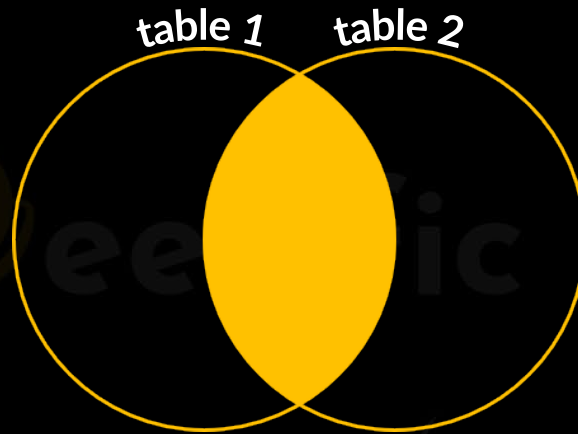
Select Result	Season	Name
	1	Cersei
	2	Tyrion
	3	Arya
	4	Cersei
	5	Tyrion

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	2

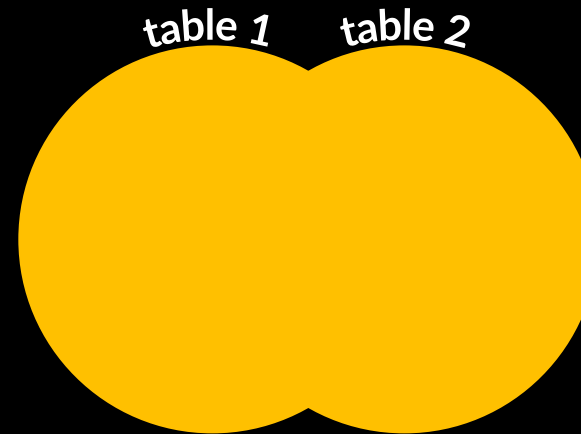


OUTER JOINS

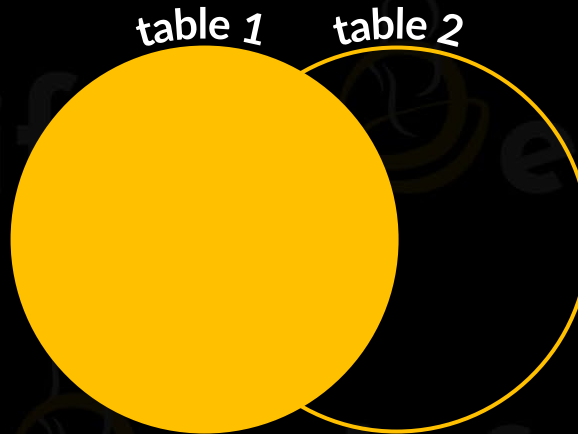
inner join



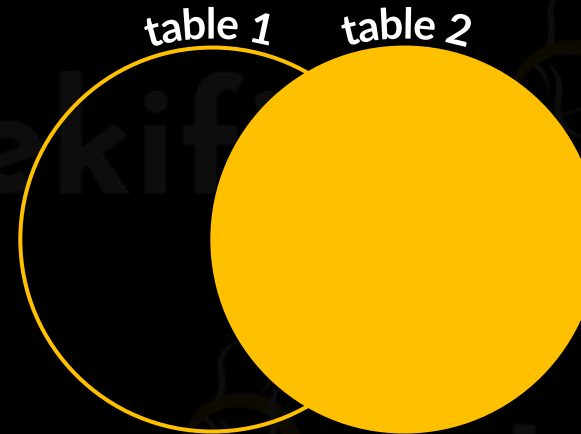
full outer
join



left outer
join



right
outer join



OUTER JOINS

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6 ?



INNER JOIN

inner join only returns rows that have a match in both tables!

```
SELECT a.Season, c.Name FROM Character AS c
       INNER JOIN Appearance AS a
       ON c.Id = a.Character
```

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

AppearanceResult	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6



LEFT OUTER JOIN

```
SELECT a.Season, c.Name FROM Character AS c
LEFT OUTER JOIN Appearance AS a
ON c.Id = a.Character
```

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Season	Name
	1	Cersei
	2	Tyrion
	3	Arya
	4	Cersei
	NULL	Sansa
	NULL	Daenerys



RIGHT OUTER JOIN

```
SELECT a.Season, c.Name FROM Character AS c
RIGHT OUTER JOIN Appearance AS a
ON c.Id = a.Character
```

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Season	Name
	1	Cersei
	2	Tyrion
	3	Arya
	4	Cersei
	5	NULL



FULL OUTER JOIN

```
SELECT a.Season, c.Name FROM Character AS c
FULL OUTER JOIN Appearance AS a
ON c.Id = a.Character
```

Appearance	Season	Character
	1	5
	2	2
	3	1
	4	5
	5	6

Character	Id	Age	Name	House
	1	11	Arya	Stark
	2	32	Tyrion	Lannister
	3	13	Sansa	Stark
	4	16	Daenerys	Targaryen
	5	36	Cersei	Lannister

Select Result	Season	Name
	1	Cersei
	2	Tyrion
	3	Arya
	4	Cersei
	5	NULL
	NULL	Sansa
	NULL	Daenerys

